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FOURTH SEMESTER B.Tech EXAMINATION, MAY-2024

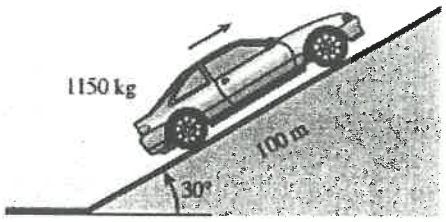
THERMAL SCIENCE

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i.** Missing data may be suitably assumed.

ii. Use of Steam Table is permitted.

ANSWER ALL QUESTIONS

		PART-A	**05 x02=10**		
1.	**a.**	Differentiate between open system and closed system.	L2	1	**02**
	b.	State the first law of thermodynamics.	L1	2	**02**
	c.	Prove $COP_{HP} = COP_R + 1$.	L2	3	**02**
	d.	Differentiate between conduction and convection.	L2	5	**02**
	e.	Define thermal boundary layer.	L1	6	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Identify the type of property of the following: **i.** Specific Volume **ii.** Weight **iii.** Entropy **iv.** Specific heat **v.** Energy	L2	1	**05**
		OR			
	b.	Obtain expression for work done for the following process: **i.** Isobaric process **ii.** Isothermal process	L3	1	**05**
3.	**a.**	Determine the power required for a 1150-kg car to climb a 100-m-long uphill road with a slope of 30° (from horizontal) in 12 s (a) at a constant velocity, (b) from rest to a final velocity of 30 m/s, and (c) from 35 m/s to a final velocity of 5 m/s. Disregard friction, air drag, and rolling resistance. 	L3	2	**05**
		OR			
	b.	Steam enters a nozzle at 400°C and 800 kPa with a velocity of 10 m/s, and leaves at 300°C and 200 kPa while losing heat at a rate of 25 kW. For an inlet area of 800 cm^2, determine the velocity and the	L3	2	**05**

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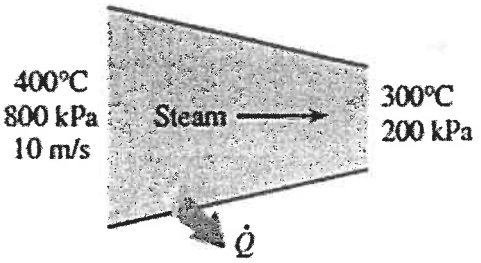
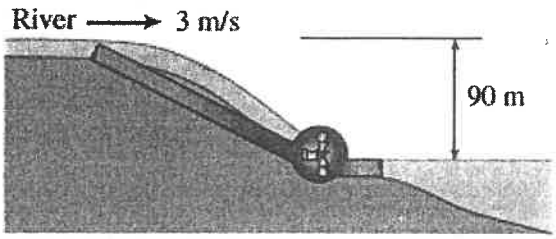
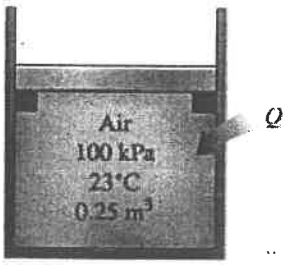
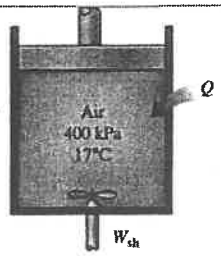
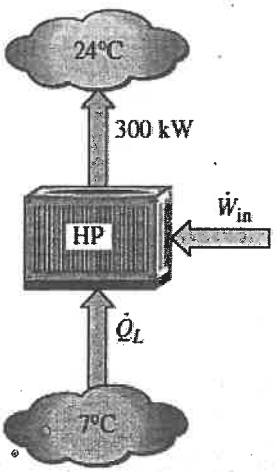
		volume flow rate of the steam at the nozzle exit. 			
4.	**a.**	A heat pump operates on a Carnot heat pump cycle with a COP of 8.7. It keeps a space at 24°C by consuming 2.15 kW of power. Determine the temperature of the reservoir from which the heat is absorbed and the heating load provided by the heat pump.	L3	3	**05**
		OR			
	b.	A Carnot heat engine receives 650 kJ of heat from a source of unknown temperature and rejects 250 kJ of it to a sink at 24°C. Determine (a) the temperature of the source and (b) the thermal efficiency of the heat engine.	L3	3	**05**
5.	**a.**	Air is compressed by a 15-kW compressor from P1to P2. The air temperature is maintained constant at 25°C during this process as a result of heat transfer to the surrounding medium at 20°C. Determine the rate of entropy change of the air. State the assumptions made in solving this problem.	L3	4	**05**
		OR			
	b.	A completely reversible heat pump produces heat at a rate of 300 kW to warm a house maintained at 24°C. The exterior air, which is at 7°C, serves as the source. Calculate the rate of entropy change of the two reservoirs and determine if this heat pump satisfies the second law according to the increase of entropy principle.	L3	4	**05**
6.	**a.**	An exterior wall of a house may be approximated by a 0.1 m layer of common brick (k = 0.7 W/m°C) followed by a 0.04 m layer of gypsum plaster (k = 0.48 W/m°C). What thickness of loosely packed rock wool insulation (k = 0.065 W/m°C) should be added to reduce the heat loss or (gain) through the wall by 80 percent?	L2	5	**05**
		OR			
	b.	How the conductive heat transfer coefficient of a material is determined? Explain.	L2	5	**05**
7.	**a.**	Illustrate thermal boundary layer.	L2	6	**05**
		OR			
	b.	Define the following non-dimensional numbers: (i) Nusselt number and (ii) Grashof number	L1	6	**05**

		PART-C **6 X 10=60**			
8.	**a.**	Consider a river flowing toward a lake at an average velocity of 3 m/s at a rate of 500 m^3/s at a location 90 m above the lake surface. Determine the total mechanical energy of the river water per unit mass and the power generation potential of the entire river at that location. 	L3	1	**10**
		OR			
	b.	Compressed helium at room temperature of 30°C from a canister is used to inflate a flexible balloon. The shape of the balloon may be considered to be spherical. The diameter of the balloon changes from 15 cm to 25 cm at the end of the inflation. If the initial pressure of He is constant at 105kPa, Determine the following: i. Mass of the gas inside the balloon ii. Work done in inflating the balloon iii. Final temperature of the gas	L3	1	**10**
9.	**a.**	Air is contained in a cylinder device fitted with a piston-cylinder. The piston initially rests on a set of stops, and a pressure of 200 kPa is required to move the piston. Initially, the air is at 100 kPa and 23°C and occupies a volume of 0.25 m^3. Determine the amount of heat transferred to the air, in kJ, while increasing the temperature to 700 K. Assume air has constant specific heats evaluated at 300 K. 	L3	2	**10**
		OR			
	b.	Air is contained in a variable-load piston-cylinder device equipped with a paddle wheel. Initially, air is at 400 kPa and 17°C. The paddle wheel is now turned by an external electric motor until 75 kJ/kg of work has been transferred to air. During this process, heat is transferred to maintain a constant air temperature while allowing the gas volume to triple. Calculate the required amount of heat transfer, in kJ/kg.	L3	2	**10**

					
10.	**a.**	A heat engine receives heat from a heat source at 1200°C and has a thermal efficiency of 40 percent. The heat engine does maximum work equal to 500 kJ. Determine the heat supplied to the heat engine by the heat source, the heat rejected to the heat sink, and the temperature of the heat sink.	L3	3	**10**
		OR			
	b.	An automobile engine consumes fuel at a rate of 22 L/h and delivers 55 kW of power to the wheels. If the fuel has a heating value of 44,000 kJ/kg and a density of 0.8 g/cm^3, determine the efficiency of this engine.	L3	3	**10**
11.	**a.**	A 30-kg aluminum block initially at 140°C is brought into contact with a 40-kg block of iron at 60°C in an insulated enclosure. Determine the final equilibrium temperature and the total entropy change for this process.	L3	4	**10**
		OR			
	b.	A completely reversible heat pump produces heat at the rate of 300 kW to warm a house maintained at 24°C. The exterior air, which is at 7°C, serves as the source. Calculate the rate of entropy change of the two reservoirs and determine if this heat pump satisfies the second law according to the increase of entropy principle. 	L3	4	**10**
12.	**a.**	A reactor's wall 320 mm thick, is made up of an inner layer of fire brick (k = 0.84 W/m°C) covered with a layer of insulation (k = 0.16 W/m°C). The reactor operates at a temperature of 1325°C and the	L3	5	**10**

		ambient temperature is 25°C. (i) Determine the thickness of fire brick and insulation which gives minimum heat loss. (ii) Calculate the heat loss presuming that the insulating material has a maximum temperature of 1200°C. If the calculated heat loss is not acceptable, then state whether addition of another layer of insulation would provide a satisfactory solution.			
		OR			
	b.	The temperature at the inner and outer surfaces of a boiler wall made of 20 mm thick steel and covered with an insulating material of 5 mm thickness are 300°C and 50°C respectively. If the thermal conductivities of steel and insulating material are 58W/m°C and 0.116 W/m°C respectively, determine the rate of flow through the boiler wall.	L3	5	**10**
13.	**a.**	A steam pipe of 10 cm ID and 11 cm OD is covered with an insulating substance k = 1 W/m K. The steam temperature is 200°C and ambient temperature is 20 °C. If the convective heat transfer coefficient between insulating surface and air is 8 W/m^2K, find the critical radius of insulation for this value of r_c. Calculate the heat loss per m of pipe and the outer surface temperature. Neglect the resistance of the pipe material.	L3	6	**10**
		OR			
	b.	A wire 1.5 mm in diameter and 150 mm long is submerged in water at atmospheric pressure. An electric current is passed through the wire and is increased until the water boils at 100°C. Under the condition if convective heat transfer coefficient is 4500 W/m2°C find how much electric power must be supplied to the wire to maintain the wire surface at 120°C?	L3	6	**10**